

Mobile App Development in the Fast-changing Mobile Phone Industry



Change is the only constant in the IT industry, and this change has been particularly dramatic in the case of the mobile phone industry. Let's take a quick look at the transformation in this industry over the last two decades, and how this has impacted mobile app development.

Mobile devices have changed drastically, starting off from Nokia's basic feature phones, moving to BlackBerry's smart/business phones, and then to the revolutionary Google and Apple smartphones. With time, hardware has improved with increasing RAM and processor capabilities. Today, there are lot of cool hardware being built into phones to make them complete devices, right from cameras and GPS to accelerometers and fingerprint readers.

Over the years, plethora of mobile OS/platforms have been created and launched by companies like Sun, Qualcomm, RIM, Microsoft, Google and Apple — right from J2ME/BREW to BlackBerry (RIM), and Android, iOS and WP8. All these platforms support different native API suites and vary in terms of the framework they offer. In fact, the base programming languages of these platforms have been different. Java was used for J2ME, RIM and Android, while C++ was used for BREW, C#/C++ for WP8 and Objective C for iOS.

With the mobile market being shared by these different platforms/OSs, till a few years back it was difficult for developers and companies to create mobile applications for all platforms. It needed a diverse set of skills to develop and maintain mobile applications for each of the mobile platforms and, thus, presented cost and time-to-market challenges.

The alternate means tried included creating hybrid apps (Web based applications wrapped in native mobile app containers). Hybrid apps had their own limitations and were deficient in terms of performance, user engagement and effective utilisation of device hardware capabilities.

The recent innovation of bot platforms has introduced

new possibilities for creating device and platform neutral apps. In the long run, bots (chat-bots) are predicted to substitute a major chunk of mobile apps. Though new, bot platforms are limited in terms of the API set, features and capabilities, as of date.

Hence, considering the diverse OS ecosystem, a mobile application design and development strategy becomes very important. A smart, strategic approach allows supporting multiple OSs/platforms and devices, without compromising on aspects like performance and user engagement.

This article is based on my years of experience in designing and developing games as well as consumer and enterprise-grade rich mobile client applications for all the earlier-mentioned platforms. Individual developers or organisations that plan to develop mobile client applications may find my experiences helpful in deciding on the most suitable mobile design and development strategy, after evaluating the pros and cons of each. I intend to share information about different mobile OSs/platforms for different era, the challenges they presented at the time, followed by the design and development strategy adopted to deal with those challenges.

Early 2000s: The era of feature phones with J2ME/BREW apps

During the early 2000s and until around 2004, BREW and J2ME games and apps were predominant in the market. The reason was simple — people were using feature phones and they needed lightweight platforms. Most feature phones supported either J2ME or BREW.